Shipment Integrity Monitor: Complete Guide and Demo Plan

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This guide explains how the shipment monitor works, from the sensors to the tamper-evident log, and gives a step-by-step script for demonstrating every feature.

The project in one page

The project is a battery-powered box that watches a shipment from sealing to delivery and keeps a timestamped, tamper-evident record of every breach. It runs on an STM32F411 (ARM Cortex-M4) Black Pill, programmed bare-metal at register level in C, with no HAL and no RTOS.

The problem. Goods in transit can be warmed, dropped, tilted or opened without anyone knowing until delivery. By then nobody can say when it happened or who is responsible. Cold-chain goods such as vaccines are the clearest case: if they leave their temperature range once, they may be unsafe even if they look fine.

The justice and compliance angle. The box produces an evidence trail. Each breach is latched, so it cannot be cleared during transit, and logged with the time. A shipper, carrier, receiver or regulator can then see whether the rules were followed and where they were broken. That protects consumers from spoiled goods and protects honest carriers from false damage claims. The hash-chained log makes it visible if anyone edits the record afterwards.

Adaptive. The same box applies different limits depending on what is inside. A medicine profile is strict about temperature, an electronics profile is strict about shock, and a produce profile tolerates more of both.

Computation. The box does more than compare readings with limits. It filters noise, predicts when a limit will be reached, scores a trip with Mean Kinetic Temperature, measures drops, tilt and vibration, fuses two shock sensors, and chains its log with a CRC so that edits are detected.

Rubric fit (23ELC302). The project uses well over four peripherals (I2C, ADC, UART, timers/PWM, GPIO interrupts, hardware CRC), has a justice/compliance topic, and has a real computing algorithm set.

How it works end to end

A trip has four stages: choose a profile, seal, monitor, deliver.

1. **Menu.** The LCD shows the profile. Button A cycles through Pharma-Cold, Electronics and Produce. Button B confirms.
2. **Wait for seal.** The LCD says "Close the lid...". When the reed switch reads closed for more than 0.5 s, the box seals.
3. **Sealing.** The firmware calibrates the accelerometer (counts per g and the reference gravity vector), records the light baseline, resets all computations, and starts the log chain with a SEALED line. In demo mode it also centres the temperature and humidity limits on the room's readings at that moment. A 3 s arming delay follows, in which readings are logged but nothing can breach.
4. **Monitoring.** About once a second the firmware runs the pipeline below. Between samples it keeps draining the accelerometer FIFO and the GPS buffer, and it reacts at once to the vibration and lid interrupts.
5. **Delivery.** Holding both buttons for 2 s writes `DELIVERED`, `SUMMARY` and `BENCH` lines, stops the accelerometer, fan and buzzer, and turns the LED blue.

The per-sample pipeline:

1. **Read** temperature and humidity (SHT31), light (LDR, median of 5), accelerometer samples from the FIFO, time (RTC), and position (GPS).
2. **Filter** the temperature with a moving average.
3. **Compute**: Mean Kinetic Temperature, trend line, time to limit, excursion time and area.
4. **Analyse motion**: vibration, shock fusion, free fall and drop height, tilt angle.
5. **Check limits** against the active profile and latch any breach as a bit that cannot clear until delivery.
6. **Act**: start the fan early if a limit is about to be reached, set the LED, sound the buzzer on a new breach, update the LCD.
7. **Log**: write one sample line, plus event lines, each ending with a sequence number and a CRC check value that covers every earlier line.
8. **Send** the line over Bluetooth to the dashboard.

Hardware

The board is an STM32F411CEU6 Black Pill (Cortex-M4F, 16 MHz internal oscillator). Everything below is wired to it and runs from a USB power bank.

Part	Interface and pins	What it does
SHT31	I2C1 (PB6 SCL, PB7 SDA), address 0x44	Temperature and humidity inside the box
ADXL345	I2C1, address 0x53	3-axis acceleration at 100 Hz, ±16 g, with FIFO. Feeds shock, drop, tilt and vibration
DS1307/DS3231 RTC	I2C1, address 0x68	Real time stamped on every log line
16x2 LCD with PCF8574	I2C1, address 0x27	Profile menu, live status pages, breach letters
LDR with 10 kΩ divider	PA0, ADC1 channel 0, 12-bit	Light inside the box, for opening detection
Reed switch and magnet	PA1, interrupt on both edges	Lid open or closed
SW-420	PB0, interrupt on both edges	Instant knock or vibration
NEO-6M GPS	USART1 (PA9, PA10), 9600 baud	Position for each log line (needs open sky for a fix)
HC-05 Bluetooth	USART2 (PA2, PA3), 9600 baud	Sends the log to the laptop dashboard
RGB LED	PB4, PB5 (TIM3), PB3 (TIM2), PWM	Green OK, amber warning, red breach, blue delivered
Buzzer	PB8	Sounds on a new breach
Relay and 5 V fan	PB9 (active-low module)	Cools the box near the upper temperature limit
Buttons	PB12 (profile), PB13 (confirm, seal, deliver)	User control

Four I2C devices share one bus, so a single stuck device can block the others. The firmware times out bus waits and flags the failed sensor (error flags for SHT31, ADXL345 or RTC) instead of hanging. Why each sensor is there: temperature and humidity protect the goods, the ADXL345 and SW-420 detect rough handling, the LDR and reed switch detect opening, the RTC and GPS say when and where, and the HC-05 gets the evidence off the box.

Firmware structure

The firmware is one bare-metal C file, `main.c`, of about 2,180 lines. It configures every peripheral through registers, with no HAL and no RTOS, and is built with ARM GCC at 16 MHz. The FPU is switched on at the start of `main()`. It compiles with zero warnings and takes about 12.5 KB of flash.

It is a super-loop with a few interrupts. The loop never blocks for long: even the delay function and the Bluetooth print routine call a background task, so the accelerometer FIFO and the GPS buffer are always serviced.

States

State	What happens	Leaves when
Menu	Button A cycles the profile, LCD shows its name	Button B confirms
Wait seal	LCD says "Close the lid..."	Lid closed for 0.5 s (seal). Button A goes back to the menu
In transit	Sampling, checks, fan, LED, log	Both buttons held for 2 s
Delivered	Final lines written, LED blue, LCD shows the result	Power cycle

Interrupts

Interrupt	Source	What the handler does
SysTick	Core timer, every 1 ms	Increments the millisecond counter used by `millis()`
EXTI0	SW-420 on PB0	Sets the shock flag and stores the time of the edge (for fusion)
EXTI1	Reed switch on PA1	Sets the lid-open flag when the lid reads open
USART1	Each GPS byte	Stores it in a 512-byte ring buffer

Handlers only set flags or store bytes. All decisions are made in the main loop, so timing stays predictable.

Background task

The background task parses GPS sentences from the ring buffer and reads the ADXL345 FIFO every 80 ms. Reading the FIFO in bursts means no 100 Hz sample is lost even while the loop is busy printing or waiting.

Robustness

- Every I2C wait has a time-out. On failure the bus is reset and the read is skipped, so a loose wire cannot freeze the box.
- A failed sensor read keeps its last good value and sets an error bit (`ERR`) in the log.
- If the LCD stops answering, writes to it are skipped and retried every 10 s.
- At boot a self-test shows `S:Y A:Y R:Y` on the LCD (SHT31, ADXL345, RTC) and reports the same over Bluetooth. A second self-test checks the hardware CRC unit and prints "CRC HW OK" or "CRC SOFTWARE" (the fallback).

Profiles

A profile is a set of limits. All sensors are read after sealing whichever profile you pick; the profile decides the limits and which checks are on.

Check	Pharma-Cold	Electronics	Produce
Temperature range (°C)	2.0 to 8.0	−10.0 to 45.0	0 to 25.0
Humidity limit (%RH)	60	85	95
Shock limit (g)	1.5	1.3	2.5
Tilt limit	45°	45°	off
Light tamper (LDR)	on	on	off
Lid open (reed switch)	on	on	on

Pharma-Cold is strict on temperature, Electronics on shock, and Produce tolerates more of both. Lid-open, GPS and vibration logging are always on.

Demo mode. A room is not a fridge, so with `DEMO_MODE` set to 1 the Pharma-Cold and Produce limits are re-centred on the room's readings at the moment of sealing. Pharma-Cold then allows 5.0 °C below to 1.5 °C above the sealing temperature and 15 %RH above

the sealing humidity. Produce allows 10.0 °C below to 8.0 °C above. Electronics keeps its real limits. The SEALED log line records the limits actually used.

Light tamper. At sealing the firmware averages eight median-filtered LDR readings as the baseline. Light tamper latches when a reading rises more than 500 ADC counts above it.

What can switch a check off. If the SHT31 fails, temperature and humidity are skipped and the fan stops being controlled. If the ADXL345 fails or cannot calibrate at sealing, tilt is skipped and the SW-420 can latch a shock on its own.

Computing

The base checks use integers (hundredths of °C, milli-g). The advanced analysis uses the Cortex-M4F's hardware floating-point unit and our own maths routines, so no maths library is linked. Every routine was tested on a PC against a reference.

Own maths routines

Routine	Method	Error vs reference
Square root	Newton iteration from a bit-trick start	1.2e-7
Exponential	Range reduction plus a degree-6 polynomial	4e-6
Natural log	atanh series	2e-6
Arctangent (degrees)	Minimax polynomial	0.0007°

1. Filtering

Temperature uses an exponential moving average with weight 1/4: new = old + (reading − old) / 4. The LDR uses the median of 5 readings, so one glitch cannot look like light. This stops noise from causing false breaches.

2. Mean Kinetic Temperature (MKT)

Degradation speeds up exponentially with temperature, so a hot hour does more harm than a cold hour undoes. MKT gives the single steady temperature with the same total effect. With ΔH/R = 10,000 K and T in kelvin:

```
MKT = \frac{\Delta H/R}{-\ln\left(\frac{1}{n}\sum_{i=1}^{n} e^{-\Delta H/(R\,T_i)}\right)}
```

The sum is taken relative to 25 °C and uses Kahan summation to keep rounding error small over long trips. A trip of 23 h at 20 °C and 1 h at 40 °C gives an MKT of 22.45 °C (reference 22.448), while the plain average is 20.83 °C. MKT is always logged and summarised, and compared with the profile's upper limit at delivery.

3. Trend and time to limit

Every second sample is stored, 60 points in a ring (about 2 minutes). A least-squares line is fitted through them using real timestamps:

```
slope = \frac{n\sum t_i T_i - \sum t_i \sum T_i}{n\sum t_i^2 - \left(\sum t_i\right)^2}
```

The fit needs at least 15 points and ignores slopes smaller than 0.10 °C/min. The time to limit is (limit − T) / slope, toward the limit the temperature is heading for. If it is under 120 s and the box is warming, the fan starts early and the LED turns amber. The test slope came out as 0.503 against a true 0.500, and a flat line gives no ETA.

4. Excursion budget

The first crossing of a limit logs `TEMP_WARNING` only. The excursion then becomes a `TEMP_EXCURSION` breach if any one of these is reached:

- 30 s of total time outside the limit (`WHY=TIME`)
- an area outside the limit of 0.50 °C·min (`WHY=AREA`)
- 3.0 °C beyond the limit (`WHY=HARD`)

This separates a brief touch from sustained abuse. The numbers are scaled for the demo; real values come from the product's stability data. Setting `EXCURSION_BUDGET` to 0 restores the plain "outside the limit is a breach" rule.

5. Accelerometer analysis

The ADXL345 runs at 100 Hz, ±16 g, in FIFO stream mode. At sealing the firmware measures counts per g and the resting gravity vector from the calmest 16 samples, so the 1 g reference is exact for that sensor.

- **Vibration.** RMS of the acceleration magnitude over a short window, logged as `VIB` in mg.
- **Shock peak.** The highest magnitude seen, logged as `PEAK`.
- **Drop.** Free fall is magnitude below 0.4 g for 4 to 60 samples (10 ms each), followed by an impact window of 10 samples. The equivalent height from the fall time t is h = ½ g t², which for 20 samples (0.2 s) is 196 mm. A drop of 10 cm or more is logged as `DROP` (with `HEIGHT_CM`, `FALL_MS`, `PEAK`) and counts as a shock breach.
- **Tilt.** The angle between the current gravity vector a and the sealed vector r:

```
\theta = \operatorname{atan2}\left(\lVert a \times r \rVert,\; a \cdot r\right)
```

Only quasi-static samples (0.8 to 1.2 g) are used, and two checks in a row must agree, so shaking is not read as tilt. Above the profile limit it logs `TILT_EXCEEDED`, and when the box is flipped it logs `INVERTED`.

6. Sensor fusion

The ADXL345 measures real acceleration but can glitch. The SW-420 is a simple switch that false-triggers. So a shock is confirmed only when both agree:

What was seen	Result
ADXL345 over limit and an SW-420 edge within 300 ms	`SHOCK_CONFIRMED`, breach
ADXL345 over limit only	`SHOCK_ACCEL`, breach
SW-420 only	`VIBRATION_SUSPECT`, logged at most once per 5 s, no breach
SW-420 only, with the ADXL345 failed	`SHOCK_SWITCH`, breach

Setting `FUSION_SW420_ALONE_LATCHES` to 1 makes an SW-420 hit alone latch a breach.

7. Integrity score

A 0 to 100 score summarises each sample from the latched breaches, with a penalty for each kind (tilt costs 15). The lowest score of the trip is kept as the worst score in the summary.

8. Cycle benchmarks

The DWT cycle counter times six operations: MKT, trend, a full computation step, a full sample, tilt, and the log CRC. The average and maximum cycles of each are printed in a `BENCH` line at delivery. Compare the cycle counts with the 16,000,000 cycles available each second at 16 MHz. They have not been measured on the board yet, so read the real numbers from your first delivery.

The tamper-evident log

The HC-05 sends one comma-separated line per sample and per event. These are real lines produced by the firmware code (on a PC), shortened for reading:

```
09/10/26 14:20:01,EVENT=SEALED,PROFILE=Pharma-
Cold,TMIN=21.5,TMAX=24.5,HMAX=70.0,GMAX=1.50,TILTLIM=45,ACAL=182,SEQ=0,CHK=ED
FB8877
14:21:00,T=24.2,H=52.4,G=1.00,LDR=1453,FIX=1,LAT=13.07270,LON=80.27070,SCORE=
100,BREACH=0,ERR=0,MKT=23.30,RATE=1.20,ETA=32,EXC=0,DEG=0.00,VIB=15,PEAK=1.02
,TILT=3,SEQ=59,CHK=BBBBDC57
14:21:21,EVENT=SHOCK_CONFIRMED,PEAK=2.15,SEQ=81,CHK=6083ED15
14:21:51,EVENT=DROP,HEIGHT_CM=19,FALL_MS=200,PEAK=4.95,SEQ=114,CHK=CED07845
14:22:32,EVENT=DELIVERED,SEQ=156,CHK=54ED261D
```

Field	Meaning
`T`, `H`	Temperature in °C and humidity in %RH
`G`, `PEAK`	Current and highest acceleration magnitude in g
`LDR`	Light reading (ADC counts)
`FIX`, `LAT`, `LON`	GPS fix flag and position
`SCORE`	Integrity score, 0 to 100
`BREACH`	Latched bits: 1 temperature, 2 humidity, 4 shock or drop, 8 light, 16 opened, 32 tilt
`ERR`	Failed sensors: 1 SHT31, 2 ADXL345, 4 RTC
`MKT`	Mean Kinetic Temperature so far, °C
`RATE`, `ETA`	Temperature trend in °C/min, and seconds to the limit (0 if none)

Field	Meaning
`EXC`, `DEG`	Seconds outside the limit, and area outside in °C·min
`VIB`, `TILT`	Vibration RMS in mg, and tilt angle in degrees
`SEQ`, `CHK`	Sequence number and chain check value

Reading BREACH as bits: the last sample of that sample run shows 37, which is 1 + 4 + 32, so temperature, shock and tilt had all latched.

How the hash chain works

After sealing, every line ends with `SEQ` and `CHK`. `CHK` is a CRC-32 (polynomial 0x04C11DB7, start value 0xFFFFFFFF) over the line's text and all the earlier lines, started with a fixed key (`LOG_KEY`). The STM32 hardware CRC unit does the work, with a software fallback that gives identical results (checked against a Python model).

- **Edit a line:** its own check value no longer matches, and every later line breaks too, because each one includes the earlier text.
- **Delete or lose a line:** the SEQ numbers have a gap, so the dashboard reports lost data instead of calling it an edit.
- **What it does not do:** someone who knows the key and the algorithm could rebuild the whole chain. It shows tampering by an ordinary user; it is not a digital signature.

The dashboard repeats the same CRC chain in JavaScript. It restarts the chain at each SEALED line and reports how many lines were checked, missing or bad, and marks the first bad line.

End-of-trip lines

`SUMMARY` gives the profile, duration, worst score, breach bits, shock and drop counts, MKT against the limit (`MKT_OK`), time and area outside the limit, tilt maximum, suspect count and the final `VERDICT` (PASS or FAIL). `BENCH` gives average and maximum CPU cycles for each algorithm.

The dashboard

Shipment Twin is a single HTML page, `shipment_twin.html`. For live use, open the laptop copy in Chrome or Edge, pair the HC-05 with the laptop, and connect to its serial port from the page (Web Serial). It can also replay a saved log. The published online copy is for viewing and the sample run; its Save log button does not work inside the viewer.

Panel	What it shows
Box twin	A picture of the box with the waybill: profile, status, and the **Log chain** cell (lines checked, missing and bad), plus the LCD mirror with its four 3-second pages
Live readings	Current temperature, humidity, shock, light and score
On-board computation	Six tiles: MKT against the limit, trend with predicted time to limit, the two excursion meters (time and area), motion (tilt box plus drop, vibration and peak), log integrity, and the cycle-cost table from `BENCH`
Trend	A chart of temperature, humidity, shock (peak), light, score or MKT over time, with breach stamps (T, H, S, L, O, I) where each latched
Event log	Every event with its time and details
Location	The GPS position
Serial feed	The raw lines as received

Buttons. Connect HC-05 opens the serial port for live use. Play sample run plays a built-in trip that includes every event, so you can demonstrate without hardware. Save log downloads the received lines. A saved log can be opened again to replay it. Demo: edit a log line lowers sample 40's temperature by 1.5 °C and reloads the log, so the chain check flags that exact line.

Sample run caution. The cycle counts in the sample run are made-up demo numbers. Only a real `BENCH` line from the board shows real costs.

Demonstration plan

The demo takes about 10 minutes and shows every upgrade in one run. Rehearse it once before the evaluation, because the timings below have not been tried on your board.

Before you start

- In `main.c` keep `DEMO_MODE` at 1, set `RTC_SET_TIME` to 1 once to set the clock, then back to 0 and flash again. `RELAY_ACTIVE_HIGH` should be 0 for an active-low relay module.
- Pair the HC-05 with the laptop and open the laptop copy of the dashboard in Chrome or Edge.
- Have ready: a hair dryer or hand warmer, a soft cushion, a small torch (phone light), and a magnet for the lid.
- Test the Play sample run button first. It is your backup if hardware misbehaves, and it shows the same events.
- Choose **Pharma-Cold**. It is the only profile with every check on. Place the box in the room where you will demo and let it settle for a minute.
- GPS needs open sky for a fix. Indoors the log shows `FIX=0`; say so rather than hiding it.

Step by step

#	Do this	You should see	Say this
1	Power on and wait for the boot screen	LCD self-test `S:Y A:Y R:Y`, and "CRC HW OK" in the serial feed	"Sensors and the hardware CRC unit pass their self-tests at boot."
2	Press A to pick Pharma-Cold, press B, close the lid and keep the box still	SEALED line with the limits and `ACAL`, then green LED	"At sealing it calibrates the accelerometer to 1 g and records its starting orientation. The log chain starts here."
3	Wait 3 s, then warm the box slowly with the dryer from a distance	`RATE` rises, `ETA` counts down on the LCD ("Limit in ..."), amber LED, fan starts before the limit	"It predicts when it will hit the limit and cools early. That is the least-squares trend line."

#	Do this	You should see	Say this
4	Let it cross the limit for a few seconds, then stop heating	`TEMP_WARNING` only, no breach	"A brief touch is only a warning. The excursion budget decides when it becomes a breach."
5	Heat it again and keep it above the limit	`TEMP_EXCURSION` with `WHY=TIME` after 30 s outside (or `AREA` / `HARD`), red LED, buzzer	"A breach needs 30 s outside, or enough area, or 3 °C beyond. These numbers are scaled for the demo."
6	Tap the box firmly	`SHOCK_CONFIRMED` with the peak g	"The accelerometer and the SW-420 agreed within 300 ms, so the shock is confirmed."
7	Flick only the SW-420 module with a finger	`VIBRATION_SUSPECT`, no new breach	"The SW-420 alone is logged but not trusted. This fusion removes false alarms."
8	Drop the box about 20 cm onto the cushion	`DROP` with `HEIGHT_CM` near 20 and `FALL_MS` near 200	"Free-fall time gives an equivalent drop height, h = ½ g t²."
9	Tilt the box beyond 45° and hold it 2 s, then turn it upside down	`TILT_EXCEEDED`, then `INVERTED`, with angles	"The angle is measured against the sealed orientation, using only still samples."
10	Optional: breathe on the SHT31, shine the torch, open the lid	`HUMIDITY_EXCURSION`, `LIGHT_TAMPER`, `UNAUTHORIZED_OPEN`	"Every breach is latched and cannot be cleared in transit."
11	On the dashboard, point to MKT, the trend and the two meters	The On-board computation panel updating	"MKT weights hot periods more than a plain average. It is the standard cold-chain measure."

#	Do this	You should see	Say this
12	Hold both buttons for 2 s	DELIVERED, SUMMARY with `VERDICT=FAIL`, BENCH, blue LED	"The box closes the record and reports the cycles each algorithm used."
13	Save the log, then replay it. Press Demo: edit a log line	Log chain shows all lines good, then flags the edited line	"Editing one line breaks its check and every one after it. The record is tamper-evident."

Tips and likely snags

- **Hold still after sealing.** Calibration needs the calmest 16 samples, and moving the box then makes the tilt reference wrong.
- **Warming.** In demo mode the upper limit is only 1.5 °C above the temperature at sealing, so a hair dryer for a few seconds is enough. The fan itself cools the box, so keep heating to stay above the limit.
- **Drop.** Free fall needs at least 4 samples (40 ms) below 0.4 g, and a 10 cm drop needs about 15. Let go cleanly, over a soft surface, with wires secured. Protect the electronics.
- **Tilt.** A shake does not count. Hold the tilted position for a couple of seconds.
- **After a breach** the box stays red until delivery. To demo again, power-cycle it and start from step 2.
- **If a feature is not seen** during the demo, show the same event in the sample run and say so.
- **Electronics profile.** Its real limit is 45 °C, so a dryer takes much longer. **Produce** has tilt and light tamper off by design.

Settings, test results and limits

Settings at the top of `main.c`

Setting	Default	Effect
`DEMO_MODE`	1	Re-centres Pharma-Cold and Produce limits on the room's readings at sealing
`RTC_SET_TIME`	0	Set to 1 once to set the clock, flash, then back to 0 and flash again
`RELAY_ACTIVE_HIGH`	0	0 for an active-low relay module
`SAMPLE_MS`	1000	Sample period
`ARM_DELAY_MS`	3000	Time after sealing in which nothing can breach
`LIGHT_DELTA`	500	ADC counts above baseline that count as light in the box
`EXCURSION_BUDGET`	1	0 restores the plain "outside the limit is a breach" rule
`EXC_TIME_BUDGET_S`	30	Seconds outside the limit before a breach
`EXC_AREA_BUDGET`	50	Area outside the limit before a breach, in 0.01 °C·min (so 0.50)
`EXC_HARD_MARGIN`	300	Distance beyond the limit for an instant breach, in 0.01 °C (so 3.0)
`PRED_FAN_S`	120	Start the fan early when the time to limit is below this
`FUSION_MS`	300	Window in which the SW-420 must agree with the accelerometer
`FUSION_SW420_ALONE_LATCHES`	0	1 lets the SW-420 alone latch a shock breach
`DROP_MIN_CM`	10	Smallest drop that counts as a shock breach

Setting	Default	Effect
`CRC_USE_HW`	1	Use the hardware CRC unit, with software fallback
`LOG_KEY`	0xA5C3E1B7	Start value of the hash chain. The dashboard uses the same key

Test results

All tests so far ran on a PC, using the firmware's own code.

- The firmware compiles with zero warnings at several optimisation levels, with hard and soft float, and links without a maths library.
- The maths routines match reference values (see the Computing section). MKT for a constant 25 °C gives 25.000.
- The accelerometer logic was run on simulated data: a 1.00 g rest reading after calibration, a 20-sample fall giving 196 mm and a 4.95 g peak, hops and sensor faults ignored, a knock detected, a 30° tilt not flagged and a 60° tilt flagged.
- The dashboard verified all 159 lines generated by the firmware code, and its edit demo flagged the exact edited line (SEQ 40). It showed no page errors in a headless browser, on desktop and phone width.

Honest limits

- **Not yet tested on the board:** the hardware CRC unit, the real ADXL345 FIFO behaviour, and the GPS fix (needs open sky). If the CRC self-test fails, the firmware uses a software CRC.
- **Tamper-evident, not tamper-proof.** Someone who knows the key and the algorithm could rebuild the chain.
- **Drop height is an equivalent height** worked out from free-fall time, not a measured height.
- **Tilt reference** is the orientation at sealing, so the box must be placed the right way up and still when sealed.
- **Budgets are demo-scaled.** Real limits must come from the product's stability data.
- **BENCH numbers** in the sample run are illustrative. Real ones come from the board.

Viva questions and short answers

- **Why bare-metal registers and not HAL?** To show control of the peripherals directly: clocks, GPIO alternate functions, I2C, ADC, UART, timers, EXTI and the CRC unit are all set by register writes.
- **Which peripherals are used?** I2C, ADC, two UARTs, timers with PWM, GPIO external interrupts, SysTick, the DWT cycle counter and the hardware CRC unit.
- **What is the justice or compliance angle?** The box gives timestamped, latched, tamper-evident evidence of how a shipment was handled, so blame and compliance can be decided from the record.
- **Where is the computing?** Filtering, Mean Kinetic Temperature, least-squares trend with time to limit, excursion budget, drop height and tilt geometry, two-sensor fusion, and a CRC hash chain.
- **Why MKT and not the average?** Chemical degradation speeds up exponentially with temperature, so a hot hour does more harm than a cold hour undoes. MKT gives the single steady temperature with the same total effect.
- **How does the prediction work?** A straight line is fitted to the last 60 stored points (about 2 minutes) by least squares. The time to limit is the distance to the limit divided by the slope. The fan starts early if it is under 120 s.
- **Why an excursion budget?** A brief touch of a limit is not the same as sustained abuse. A warning comes first, and a breach needs time, area, or distance beyond the limit.
- **Why fuse two shock sensors?** The ADXL345 measures real acceleration but can glitch. The SW-420 is a switch that false-triggers. Requiring both within 300 ms removes most false alarms.
- **How is drop height found?** From free-fall time: the accelerometer reads under 0.4 g while falling, the number of 10 ms samples gives t, and $h = \frac{1}{2} g t^2$. It is an equivalent height.
- **How is tilt found?** As the angle between the sealed gravity vector and the current one, using atan2 of the cross product and dot product, on still samples only.
- **Why a hash chain?** Each line's check value depends on every line before it, so editing or removing one line breaks all later checks. It shows tampering; it does not prevent it.
- **Why not use a maths library?** To keep the code small and to show the algorithms. Square root, exponential, log and arctangent are written by hand and tested against reference values.
- **Why the arming delay?** Moving and closing the box would otherwise cause false breaches right after sealing.

- **What happens if a sensor fails?** Its checks are skipped, an error bit is set in the log, and the rest keeps running. The SW-420 can then latch a shock on its own if the accelerometer is the failed one.
- **What would you change for a real product?** Use real stability data for the budgets, a digital signature in place of the CRC chain, a cellular modem, and a battery-backed log in flash.